Benchmarking Machine Learning Models for Road-Level Flood Prediction and Flood-Aware Routing: A Case Study of Da Nang, Vietnam

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***Abstract*—During heavy rain, drivers need to know which roads are flooded, yet flood studies in Vietnamese cities map susceptibility over areas, and flood models are compared only by classification scores. It is therefore unknown whether the model that best predicts flooded roads also gives the safest routes. This paper benchmarks six machine learning models from five families—logistic regression, support vector machine, random forest, XGBoost, multilayer perceptron, and the graph neural network GraphSAGE—for road-level flood prediction and for the flood-aware routing built on their predictions, using the extreme rainfall of 14 October 2022 in Hai Chau, Da Nang, Vietnam, as a case study. Under spatial block cross-validation over [need results] road segments, [need results: best classifier] achieved the highest area under the precision–recall curve ([need results]), while random cross-validation overstated AUC-ROC by [need results: range]. Used as routing costs, the out-of-fold probabilities of [need results: best routing model] reduced exposure to flooded segments by [need results]% at a median travel-time increase of [need results]%, and [need results: whether the best classifier was also the best for routing]. Shapley additive explanations identified [need results: top features] as the main flood drivers, and [need results: one-sentence efficiency and robustness result].**

***Index Terms*—benchmarking, explainable artificial intelligence, flood-aware routing, graph neural networks, machine learning, road networks, spatial cross-validation, urban flooding**

I. Introduction

Urban flooding disrupts road transport long before it damages buildings. A road becomes unsafe for most vehicles at about 30 cm of standing water [1], and the disruption spreads beyond the flooded links. During Hurricane Harvey, flooding of only 1.3% of the roads in Houston increased network-wide travel times by 8% [2]. For drivers and traffic managers, the useful question during heavy rain is therefore which road segments become impassable, not which areas are susceptible.

Da Nang, Vietnam, illustrates this problem. On the night of 14 October 2022, the city received close to 600 mm of rain; parts of the city stood about 1 m deep, and the Dien Bien Phu underpass flooded completely [3], [4]. Hai Chau, the dense central district along the Han River, is classed among the most flood-susceptible areas of the city [5].

Flood research in Da Nang maps susceptibility with machine learning [5], [6], simulates the sewer network [7], and maps inundation from radar imagery [8], but it reports results for areas, grid cells, or sewer nodes rather than roads. Three gaps remain. First, road-level prediction is rare, and the closest study [9] validated its models with random splits, which overstate accuracy [10]. Second, model comparisons cover mostly tree ensembles [5], [6], [9], [11], so linear, kernel, neural, and graph models are rarely tested on the same data. Third, flood models are judged by classification scores alone. This leaves open the question that this paper asks: *does the model that best predicts flooded roads also give the safest routes?*

To answer it, this paper benchmarks six machine learning models from five families for road-level flood prediction and flood-aware routing in Hai Chau during the 2022 Da Nang flood. Its contributions are as follows:

1) **Leakage-aware classification benchmark:** six models compared under spatial block cross-validation, with precision–recall, calibration, significance tests, and SHAP.
2) **Downstream routing benchmark:** the out-of-fold probabilities of every model used as routing costs and compared with shortest-path and oracle baselines.
3) **Classification–routing agreement:** a test of whether the two model rankings agree.
4) **Efficiency and robustness:** the cost of each model, a feature-group ablation, and a sensitivity analysis.

Section II reviews related work, Section III describes the case study, Section IV formulates the problem and the benchmark, Section V reports the results, Section VI discusses them, and Section VII concludes the paper.

II. Related Work

A. Flood Studies in Da Nang

Nguyen et al. [5] compared five tree ensembles across the city (AUC 0.983–0.989) and placed Hai Chau in the high and very-high susceptibility classes, and Hoang et al. [6] trained XGBoost on 1,000 flood points from media reports (accuracy 0.93). Both used random train–test splits, which overstate accuracy when nearby samples are correlated [10]. Doan et al. [7] simulated the 2022 rainfall with MIKE Urban and found that riverside manholes flooded when high river levels blocked the outfalls, and Nguyen and Quach [8] mapped the October 2022 flooding from Sentinel-1 imagery, mostly outside the dense urban core.

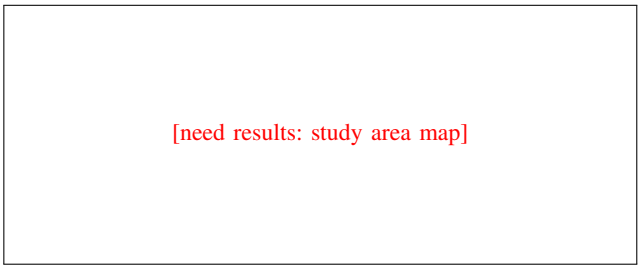


Fig. 1. Study area, road network, and observed flooded segments in Hai Chau.

B. Road-Level Flood Prediction

Ha et al. [11] mapped flash-flood susceptibility along a highway in Hoa Binh, and Sadler et al. [12] modeled street flood severity from crowdsourced reports. Yuan et al. [9] are closest to this work: random forest predicted flooded roads in Harris County, Texas, from topographic, hydrologic, and precipitation features (AUC 0.86 for Hurricane Harvey), but under random cross-validation. Pan et al. [13] used a Bayesian network to place road flood sensors in New York City; their work observes flooding on segments, whereas this paper predicts it. Graph neural networks such as GraphSAGE [14] can use the connectivity between segments, but they have not been compared with feature-based models for this task. [To do: summarize [15] and [16].]

C. Routing Under Disruption

Le et al. [17] optimized urban bus routes with multi-objective deep reinforcement learning, without hazard costs. Linking segment-level flood probabilities to routing connects these two lines of work.

As Table I shows, [need results: after verifying the table: no prior study combines road-level prediction, spatial validation, calibration, and a routing test.]

III. Case Study

A. Study Area and Historical Flood Event

The study covers the former Hai Chau district of Da Nang, with boundaries as they stood in October 2022, on the west bank of the Han River. The district contains the commercial core of the city, a dense street grid, and the Dien Bien Phu underpass, and it is classed among the most flood-susceptible areas of the city [5].

The historical event studied is the extreme rainfall of 14–15 October 2022. Da Nang received close to 600 mm of rain overnight, parts of the city stood about 1 m deep, and the Dien Bien Phu underpass flooded completely [3], [4]. Manholes along the Han River also overflowed, because high river levels blocked the drainage outfalls [7]. [need results: station rainfall totals and peak hourly intensity, 14–15 October 2022.] Fig. 1 shows the study area.

B. Road Network and Flood Labels

The drivable network was obtained from OpenStreetMap with OSMnx [18] and split at intersections and at a maximum length of 100 m, giving [need results] nodes, [need results] directed edges, and [need results] segments. The flood labels come from verified reports of flooding during the event, cross-checked against Sentinel-1 flood extents [8], following the report-based approach of Hoang et al. [6]. A segment is labeled flooded ($y = 1$) if flooding was observed on it and non-flooded ($y = 0$) otherwise. Segments within [need results: buffer distance] of a flooded segment but without their own observation are treated as uncertain and excluded; a sensitivity run that keeps them as non-flooded is reported in Section V-I. The benchmark uses [need results] flooded, [need results] non-flooded, and [need results] excluded segments.

C. Features

Each segment is described by features computed within a 25–50 m buffer. Terrain features are the minimum and mean elevation, slope, and topographic position index from a [need results: resolution] m digital elevation model (DEM). Hydrological features are the distances to the Han River and to the nearest canal, retention pond, and drainage outfall. Imperviousness is the mean normalized difference built-up index (NDBI) from Sentinel-2 imagery. Network features are road class, length, node degree, and betweenness centrality.

IV. Problem Formulation and Benchmark Design

A. Problem Formulation

The road network is a directed graph $G = (V, E)$, where V is the set of intersections and E is the set of road segments. Each segment $e_i \in E$ has a feature vector $\mathbf{x}_i \in \mathbb{R}^d$, a length ℓ_i, a free-flow travel time t_i, and a flood label $y_i \in \{0, 1\}$. The benchmark has two tasks.

1) Classification task: A model estimates the probability that segment e_i floods:

$$p_i = P(y_i = 1 \mid \mathbf{x}_i). \tag{1}$$

For the graph model, the input also includes the features of neighboring segments, taken from the line graph $L(G)$, whose nodes are the segments of G and whose edges join segments that share an intersection.

2) Routing task: The probabilities are converted into edge costs

$$C_i = t_i\,(1 + \lambda\, p_i), \tag{2}$$

where $\lambda \geq 0$ controls risk aversion. For an origin–destination (OD) pair (o, d) with the set of paths Π_{od}, the flood-aware route is

$$\pi^* = \arg\min_{\pi \in \Pi_{od}} \sum_{e_i \in \pi} C_i. \tag{3}$$

A route π is evaluated by its observed flood exposure and its travel-time increase over the shortest-time route π_0:

$$F(\pi) = \sum_{e_i \in \pi} y_i\, \ell_i, \qquad \Delta T(\pi) = \frac{T(\pi) - T(\pi_0)}{T(\pi_0)}, \tag{4}$$

where $T(\pi) = \sum_{e_i \in \pi} t_i$. The research question is whether the model with the best classification score also gives the lowest F at a given ΔT.

TABLE I
SUMMARY OF RELATED FLOOD PREDICTION AND ROUTING STUDIES

Study	Area	Prediction unit	Models compared	Spatial validation	Calibration	Routing tested
Nguyen et al. [5]	Da Nang	Area	Five tree ensembles	×	[NR]	×
Hoang et al. [6]	Da Nang	Point	XGBoost	×	[NR]	×
Ha et al. [11]	Hoa Binh	Road corridor	[NR]	[NR]	[NR]	×
Sadler et al. [12]	[NR]	Street	Poisson regression, RF	[NR]	[NR]	×
Yuan et al. [9]	Harris County	Road	RF, [NR]	×	[NR]	×
Pan et al. [13]	New York City	Expressway	Bayesian network	[NR]	[NR]	×
Le et al. [17]	Da Nang	Route	Deep RL	–	–	✓[a]
This paper	Da Nang (Hai Chau)	Road segment	Six models, five families	✓	✓	✓

[a]Routing without flood or hazard costs. [NR]: to be verified against the cited paper.

B. Models for Comparison

Six models from five families are compared:

- **Linear:** logistic regression (LR), a transparent baseline.
- **Kernel:** support vector machine (SVM) with a radial basis function kernel [19] and Platt scaling.
- **Tree ensembles:** random forest (RF) [20] and XGBoost [21].
- **Neural:** multilayer perceptron (MLP) with [need results: architecture].
- **Graph:** GraphSAGE [14] on the line graph $L(G)$, with [need results: layers and aggregator].

All models receive the same features, folds, and random seeds. Class weights handle class imbalance, features are standardized for all models except the tree ensembles, and hyperparameters are tuned by [need results: search method] inside the training folds only.

C. Classification Benchmark

Neighboring segments share terrain and drainage, so random splits leak information between training and test sets [10]. The district is therefore divided into square blocks of [need results: block size] m, which are assigned to five folds and held out in turn. For GraphSAGE, held-out segments stay in the graph, but their labels are hidden during training. Random five-fold cross-validation is also run, and the optimism gap of model m is

$$\Delta_m = \mathrm{AUC}_m^{\mathrm{random}} - \mathrm{AUC}_m^{\mathrm{spatial}}. \tag{5}$$

The primary metric is the area under the precision–recall curve (AUC-PR), whose no-skill baseline equals the share of flooded segments ([need results]). AUC-ROC, F1, and recall on the flooded class are also reported, and calibration is measured with the Brier score [22], because the probabilities become routing costs. Differences across folds are tested with the Friedman and Wilcoxon signed-rank tests [23]. SHAP values [24] are computed for every model, with TreeSHAP for RF and XGBoost and KernelSHAP on a sample of [need results: sample size] segments for the others, and the models are compared by the rank of each feature's mean absolute SHAP value.

D. Routing Benchmark

Every segment receives the probability predicted in the fold where it was held out, so no model routes on labels it was trained on. For [need results: number of pairs] random OD pairs, routes are computed with Dijkstra's algorithm for each model and for two baselines: shortest time ($p_i = 0$) and an oracle ($p_i = y_i$) that knows the observed labels. Each route is scored by its flooded segments and length, its travel-time increase, and whether a flood-free route exists. The main results use $\lambda =$ [need results], and λ is varied over [need results: range] so that models can be compared at equal travel-time cost. Finally, the model rankings by classification and by routing are compared.

E. Ablation, Sensitivity, and Efficiency

The feature-group ablation removes the terrain, hydrological, imperviousness, or network features in turn. For GraphSAGE, the number of layers is varied, and a version without graph edges isolates the effect of connectivity. Sensitivity is tested for the block size, the handling of uncertain segments, and λ. Efficiency is measured by the number of parameters, the training time, the inference time for the district, and the routing time per OD pair, on [need results: hardware and software]. All experiments use [need results: number of seeds] seeds and report the mean $\pm$ standard deviation.

V. RESULTS

A. Flood Labels

Of the [need results] segments, [need results] ([need results]%) were observed flooded and [need results] were excluded as uncertain. The flooded segments cluster [need results: spatial pattern, e.g., along the Han River or near the underpass].

B. Classification Benchmark

Table II compares the six models, and Fig. 2 shows their ROC and precision–recall curves under spatial cross-validation. [need results: name the best model by AUC-PR and recall, and how far apart the models are.] [need results: Friedman test result and which pairwise Wilcoxon differences are significant.]

TABLE II
CLASSIFICATION BENCHMARK UNDER SPATIAL BLOCK CROSS-VALIDATION (MEAN ± STANDARD DEVIATION OVER FIVE FOLDS)

		Spatial CV					Random CV
Model	Family	AUC-PR	AUC-ROC	F1	Recall (flooded)	Brier score ↓	AUC-ROC
Logistic regression	Linear	[NR]	[NR]	[NR]	[NR]	[NR]	[NR]
SVM (RBF)	Kernel	[NR]	[NR]	[NR]	[NR]	[NR]	[NR]
Random forest	Tree ensemble	[NR]	[NR]	[NR]	[NR]	[NR]	[NR]
XGBoost	Tree ensemble	[NR]	[NR]	[NR]	[NR]	[NR]	[NR]
MLP	Neural	[NR]	[NR]	[NR]	[NR]	[NR]	[NR]
GraphSAGE	Graph	[NR]	[NR]	[NR]	[NR]	[NR]	[NR]

Best value per column in bold. ↓: lower is better. [NR]: need results.

[need results: ROC and precision–recall curves of the six models]

Fig. 2. ROC (left) and precision–recall (right) curves under spatial block cross-validation. The dashed line in the right panel is the no-skill baseline.

[need results: AUC-ROC under random vs spatial CV per model]

Fig. 3. AUC-ROC of each model under spatial block and random cross-validation. Labels give the optimism gap Δ_m.

[need results: road risk map of the best model]

Fig. 4. Out-of-fold flood probability per road segment from the best model, in three risk classes.

[need results: SHAP summary plot]

Fig. 5. SHAP summary plot of the best model.

C. Optimism of Random Cross-Validation

Fig. 3 compares AUC-ROC under random and spatial cross-validation. [need results: range of the optimism gap Δ_m, which model is inflated most, and whether the model ranking changes between the two protocols.]

D. Risk Map and Feature Effects

Fig. 4 maps the out-of-fold flood probability of each segment for the best model. [need results]% of the segments fall in the high-risk class, concentrated in [need results: area]. Fig. 5 summarizes the SHAP values of the best model. [need results: top three features and the direction of their effects.] Fig. 6 compares the feature rankings of all six models. [need results: whether the six models agree on the main flood drivers, and which models differ, e.g., GraphSAGE relying more on network features.]

E. Feature-Group Ablation

Table III reports the change in performance when each feature group is removed, and the effect of the graph structure in GraphSAGE. [need results: which feature group matters most, and whether connectivity adds information beyond the segment features.]

F. Downstream Routing

Table IV compares the routing results of the six models with the two baselines, and Fig. 7 plots flood exposure against travel-time increase. [need results: which model gives the fewest flooded segments, how close it comes to the oracle, and at what travel-time cost.] Fig. 8 shows the trade-off curve of each model as λ varies. [need results: which model is best at equal travel-time cost.] [need results: whether the ranking by routing matches the ranking by AUC-PR in Table II, and which classification metric best explains the routing ranking, e.g., recall or Brier score.]

G. Representative Route

Fig. 9 shows the routes of one representative OD pair, [need results: origin and destination, e.g., a trip crossing the Dien Bien Phu underpass], and Table V lists their metrics. [need results: how the flood-aware routes avoid the flooded segments and at what detour.]

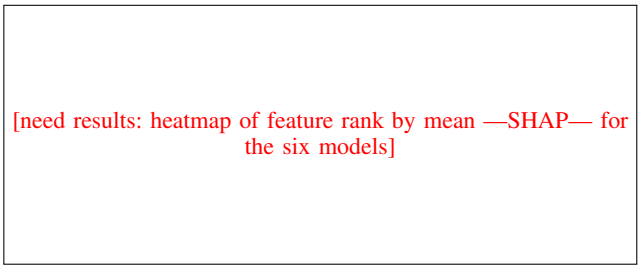


Fig. 6. Rank of each feature by mean absolute SHAP value for the six models (1 = most important). TreeSHAP for RF and XGBoost; KernelSHAP on a sample of [NR] segments for the other models.

TABLE III
FEATURE-GROUP AND GRAPH ABLATION (AUC-PR UNDER SPATIAL CV)

Setting	Model A	Model B	ΔAUC-PR
All features	[NR]	[NR]	–
− Terrain	[NR]	[NR]	[NR]
− Hydrological	[NR]	[NR]	[NR]
− Imperviousness	[NR]	[NR]	[NR]
− Network	[NR]	[NR]	[NR]
GraphSAGE, no graph edges	–	[NR]	[NR]
GraphSAGE, [NR] layers	–	[NR]	[NR]
GraphSAGE, [NR] layers	–	[NR]	[NR]

Model A, B: [NR] (e.g., XGBoost and GraphSAGE). Δ: change for model B.

H. Computational Efficiency

Table VI reports the size and run time of each model. [need results: which model is fastest and slowest, and whether the gain of the slower models justifies their cost.]

I. Sensitivity Analysis

Fig. 10 and Table VII show how the results change with the spatial block size, the handling of uncertain segments, and λ. [need results: whether the model ranking stays stable across these settings.]

VI. Discussion

A. Classification Scores and Routing Decisions

[need results: explain why the best classifier was or was not the best for routing, e.g., missed flooded segments (low recall) or poorly calibrated probabilities change the routes more than AUC suggests.] For practitioners, the benchmark indicates that models for flood-aware routing should be selected by their routing outcomes, or at least by recall and calibration, not by AUC alone.

B. Flood Mechanisms

[need results: interpret which mechanisms the models recover, e.g., tide-locked outfalls near the river, as reported in [7].]

C. Cost of Model Complexity

[need results: whether the graph and neural models gain enough over the tree ensembles to justify their training and inference cost, using Tables III and VI.]

TABLE IV
ROUTING RESULTS OVER [NR] OD PAIRS (λ = [NR])

Strategy	Flooded seg. per trip	Flooded length (m)	Time increase (median, %)	No safe route (%)
Shortest time	[NR]	[NR]	0	–
LR	[NR]	[NR]	[NR]	[NR]
SVM (RBF)	[NR]	[NR]	[NR]	[NR]
Random forest	[NR]	[NR]	[NR]	[NR]
XGBoost	[NR]	[NR]	[NR]	[NR]
MLP	[NR]	[NR]	[NR]	[NR]
GraphSAGE	[NR]	[NR]	[NR]	[NR]
Oracle (observed)	[NR]	[NR]	[NR]	[NR]

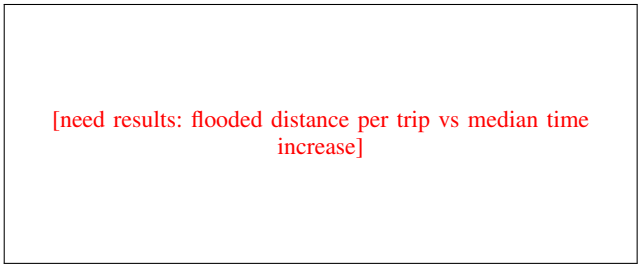


Fig. 7. Flood exposure per trip versus median travel-time increase for each strategy at λ = [NR].

D. Limitations

Four limitations apply. First, the study uses one event, so it identifies flood-prone segments rather than forecasting floods for any storm. Second, unreported flooding outside the excluded buffer is still labeled as non-flooded, which lowers the measured recall. Third, rainfall varies little across a single district, so it cannot separate segments within one event. Fourth, a coarse DEM may miss street-scale depressions. [need results: further limitations found in the results, e.g., possible reporting bias toward long or busy roads.]

VII. Conclusion

This paper benchmarked six machine learning models from five families for road-level flood prediction and flood-aware routing during the 2022 Da Nang flood. [need results: best classifier] reached the highest AUC-PR of [need results] under spatial cross-validation, and [need results: best routing model] reduced flood exposure by [need results]% for a [need results]% increase in travel time. [need results: one sentence on efficiency and robustness.]

Future work will add further events, timestamped depth records for short-term prediction, and risk costs in reinforcement-learning routing [17]. The risk maps could also provide prior flood probabilities for placing road sensors or cameras in Da Nang, following [13].

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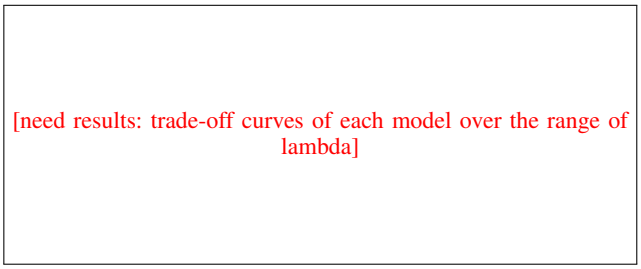


Fig. 8. Trade-off between flood exposure and travel-time increase as λ varies, for each model and the oracle.

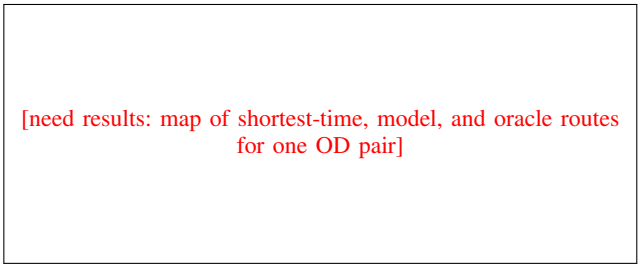


Fig. 9. Routes for a representative OD pair on the risk map: shortest time, [need results: selected models], and oracle.

TABLE V
METRICS OF THE REPRESENTATIVE ROUTE

Strategy	Travel time (min)	Flooded segments	Flooded length (m)
Shortest time	[NR]	[NR]	[NR]
[need results: model]	[NR]	[NR]	[NR]
[need results: model]	[NR]	[NR]	[NR]
Oracle (observed)	[NR]	[NR]	[NR]

TABLE VI
MODEL SIZE AND RUN TIME

Model	Parameters	Training time (s)	Inference, district (ms)	AUC-PR
LR	[NR]	[NR]	[NR]	[NR]
SVM (RBF)	[NR]	[NR]	[NR]	[NR]
RF	[NR]	[NR]	[NR]	[NR]
XGBoost	[NR]	[NR]	[NR]	[NR]
MLP	[NR]	[NR]	[NR]	[NR]
GraphSAGE	[NR]	[NR]	[NR]	[NR]

Routing time per OD pair: [NR] ms for all models.

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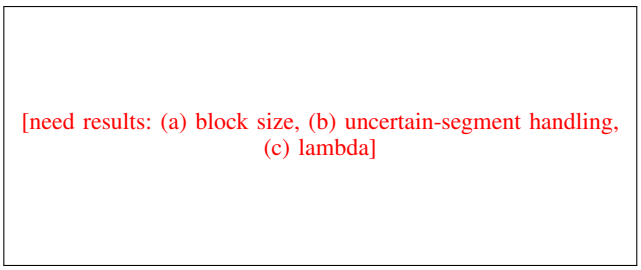


Fig. 10. Sensitivity of the results to (a) the spatial block size, (b) the handling of uncertain segments, and (c) the risk-aversion weight λ.

TABLE VII
SENSITIVITY OF THE BEST MODEL TO THE VALIDATION AND LABELING SETTINGS

Setting	AUC-PR	Recall	Brier ↓
Block size [NR] m	[NR]	[NR]	[NR]
Block size [NR] m (main)	[NR]	[NR]	[NR]
Block size [NR] m	[NR]	[NR]	[NR]
Uncertain segments excluded (main)	[NR]	[NR]	[NR]
Uncertain segments as non-flooded	[NR]	[NR]	[NR]